THE UNIVERSAL PROPERTY OF THE FIRST ORDER INFINITESIMAL NEIGHBOURHOOD OF AN IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-morphisms.tex; no statement or proof step was rewritten, reordered or removed. The author writes 'We will only prove (1)'; that restriction is part of the author's text and is reproduced as written.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-more-morphisms/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; the Xy-pic xy environment is required because the counted statement displays two commutative squares. No macro was invented or redefined.

Editorial note. The cross-chapter standard prerequisite is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- The n-th order infinitesimal neighbourhood Z_n of an immersion Z -> X. Same chapter, Definition environment labelled definition-first-order-infinitesimal-neighbourhood (source line 684, Stacks tag 04EY); the proof uses the open U of that construction and the ideal I cutting out Z in U. Supplies the objects Z' and Z_n together with the ideal I and the fact that Z is closed in U; the factorisation argument is the counted content.
- Criterion for a morphism to factor through the closed subscheme cut out by an ideal. Schemes, Lemma labelled schemes-lemma-characterize-closed-subspace (canonical Stacks tag 01HP); invoked at proof lines 739-741. Supplies the criterion that a morphism factors through the closed subscheme cut out by I as soon as the pullback of I becomes zero; the verification of that vanishing is retained in the counted span.

Source reference. Original lemma 01HP: sources/stacks-chow/tags.txt, source lines 1950–1950. Cross-chapter standard prerequisite invoked once at proof lines 739-741 for the factorisation criterion through a closed subscheme cut out by an ideal; the reduction to the vanishing b^#(b^{-1}(I^2)) = 0 is retained verbatim in the counted span.

Lemma 0.1. *Let $i : Z \to X$ be an immersion of schemes.*

(1) *The first order infinitesimal neighbourhood Z' of Z in X has the following universal property: Given any commutative diagram*

$$\begin{array}{ccc} Z & \xleftarrow{a} & T \\ {\scriptstyle i}\downarrow & & \downarrow \\ X & \xleftarrow{b} & T' \end{array}$$

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where $T \subset T'$ is a first order thickening over X, there exists a unique morphism $(a', a) : (T \subset T') \to (Z \subset Z')$ of thickenings over X.

(2) *For $n \geq 1$ the nth order infinitesimal neighbourhood Z_n of Z in X has the following universal property: Given any commutative diagram*

$$\begin{array}{ccc} Z & \xleftarrow{a} & T \\ {\scriptstyle i}\downarrow & & \downarrow \\ X & \xleftarrow{b} & T' \end{array}$$

where $T \subset T'$ is an nth order thickening over X, there exists a unique morphism $(a', a) : (T \subset T') \to (Z \subset Z_n)$ of thickenings over X.

Proof. We will only prove (1). Let $U \subset X$ be the open used in the construction of Z', i.e., an open such that Z is identified with a closed subscheme of U cut out by the quasi-coherent sheaf of ideals $\mathcal{I}$. Since $|T| = |T'|$ we see that $b(T') \subset U$. Hence we can think of b as a morphism into U. Let $\mathcal{J} \subset \mathcal{O}_{T'}$ be the ideal cutting out T. Since $b(T) \subset Z$ by the diagram above we see that $b^\sharp(b^{-1}\mathcal{I}) \subset \mathcal{J}$. As T' is a first order thickening of T we see that $\mathcal{J}^2 = 0$ hence $b^\sharp(b^{-1}(\mathcal{I}^2)) = 0$. By Schemes, Lemma 01HP this implies that b factors through Z'. Denote $a' : T' \to Z'$ this factorization and everything is clear. □